

A Comparative Study of Semantic Similarity, Rubric-Based, Generative AI, and Hybrid Scoring Methods for Automated Essay Evaluation

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Abstract— Essay answer assessment is inherently subjective, particularly in large-scale classes. Although recent advances in artificial intelligence have enabled various automated essay scoring methods, systematic empirical comparisons, especially involving hybrid strategies, remain limited. This study comparatively evaluates four approaches: Semantic Similarity Scoring, Rubric-Based Automated Scoring, Generative AI Scoring, and a Hybrid Scoring method that integrates these techniques. Automated scores are benchmarked against instructor-assigned grades using Pearson correlation, Mean Absolute Error (MAE), and the Intraclass Correlation Coefficient (ICC). The results indicate that each method exhibits distinct strengths and limitations. Semantic similarity captures semantic alignment but is less sensitive to completeness, rubric-based scoring provides more structured evaluation, and generative AI scoring offers contextual assessment with higher variability. The proposed hybrid approach achieves more balanced performance, improving agreement with instructor grades and reducing scoring error. These findings suggest that hybrid automated scoring can serve as a reliable decision-support tool for instructors rather than a replacement for human judgment.

Keywords—essay scoring, automated essay scoring, semantic similarity, rubric-based scoring, generative AI, hybrid scoring

I. INTRODUCTION

Essay assessment plays a crucial role in evaluating students' conceptual understanding, analytical ability, and critical thinking skills. Compared to objective tests, essay questions allow deeper and more expressive reasoning. However, manual grading is often affected by subjectivity, assessor fatigue, and scalability issues, particularly in large classes, which may reduce scoring consistency and reliability.

To address these challenges, Automated Essay Scoring (AES) methods have been developed using advances in artificial intelligence and natural language processing. Semantic Similarity Scoring evaluates the alignment between student responses and reference answers using vector-based text representations, but it tends to overlook structural quality and argument depth. Rubric-Based Automated Scoring incorporates explicit academic criteria to improve transparency and consistency, although it may struggle with complex semantic context and diverse writing styles. More recently, Generative AI Scoring based on Large Language Models (LLMs) has enabled holistic and context-aware evaluation, yet often exhibits higher score variability when applied independently.

Given the complementary strengths and limitations of these approaches, hybrid scoring strategies have emerged to achieve more balanced and stable assessment outcomes. However, empirical studies that systematically compare semantic similarity, rubric-based, generative AI, and hybrid

scoring approaches within a unified evaluation framework remain limited. Therefore, this study evaluates and compares the performance of Semantic Similarity Scoring, Rubric-Based Automated Scoring, Generative AI Scoring, and Hybrid Scoring by benchmarking their outputs against instructor-assigned grades using statistical performance metrics..

II. LITERATURE REVIEW

Automated Essay Scoring (AES) has attracted growing attention as a computational solution for large-scale text assessment due to the limitations of manual grading in terms of subjectivity, scalability, and consistency [1], [2]. Early AES approaches relied on surface-level textual features such as sentence length, word frequency, and syntactic complexity [3], but were criticized for their inability to capture semantic meaning and conceptual quality [4]. This limitation motivated the adoption of semantic representation-based methods.

Semantic Similarity Scoring measures semantic alignment between student responses and reference answers using vector-based representations, including Latent Semantic Analysis (LSA) [5], word embeddings [6], and Transformer-based models such as BERT [7], [8]. While effective in capturing conceptual similarity despite lexical variation, these methods primarily assess global semantic overlap and remain less sensitive to argument structure, reasoning depth, and content completeness [9], [10].

To improve interpretability and consistency, Rubric-Based Automated Scoring integrates explicit academic criteria into AES systems, enabling more structured and transparent evaluation aligned with conventional grading practices [11]–[13]. However, rubric-based approaches may struggle to capture implicit reasoning and stylistic diversity not explicitly defined in rubric criteria [14]–[16]. More recently, Generative AI Scoring based on Large Language Models (LLMs) has enabled holistic and context-aware evaluation, showing strong agreement with human graders in assessing coherence and reasoning quality [17]–[19]. Nevertheless, challenges related to score variability, prompt sensitivity, and reproducibility limit its reliability as a standalone method [20]–[22].

To address these limitations, hybrid AES strategies that integrate semantic similarity, rubric-based scoring, and generative AI evaluation have been proposed [23], [24]. Prior studies suggest that hybrid approaches yield more stable performance and higher agreement with human assessment by balancing semantic alignment, rule-constrained evaluation, and contextual reasoning [25]. However, systematic empirical comparisons within a unified experimental framework remain limited and task-dependent [26], [27]. Accordingly, this study

conducts a comparative evaluation of semantic similarity, rubric-based, generative AI, and hybrid scoring methods using statistical performance metrics benchmarked against instructor-assigned grades..

III. RESEARCH METHODOLOGY

A. Essay Scoring System Workflow

This study employs a quantitative experimental approach with a comparative–evaluative design to assess the performance of three AI-based essay scoring methods—Semantic Similarity Scoring, Rubric-Based Automated Scoring, and Generative AI Scoring—by benchmarking their outputs against instructor-assigned grades as the ground truth. In this framework, artificial intelligence functions as a decision-support system that provides scoring recommendations while preserving the instructor’s authority as the final evaluator.

Table 1 Essay Scoring Methods Used in This Study

Method	Scoring Principle	Evaluation Focus	Output
Semantic Similarity Scoring	Measures semantic similarity between student responses and reference answers using text embeddings and cosine similarity	Semantic alignment	Semantic score (0–100)
Rubric-Based Automated Scoring	Automated evaluation based on explicit academic rubrics (concept accuracy, content coverage, coherence)	Structured academic quality	Rubric score (0–100)
Generative AI Scoring	Direct evaluation by a Large Language Model (LLM) as a holistic assessor without explicit rubric constraints	Generative AI assessment	LLM score (0–100)
Hybrid Scoring	Linear combination of semantic and rubric-based scores for integrated analysis	Comparative reference	Hybrid score

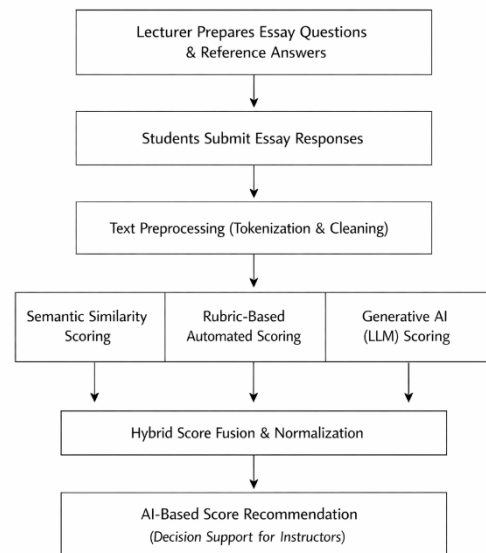


fig. 1 Architecture of the proposed hybrid essay scoring system

Student responses are evaluated in parallel by multiple AI modules and integrated to produce a hybrid score for instructor decision support.

B. Semantic Similarity Scoring

The dataset consists of instructor-designed essay questions, instructor-provided reference answers, and student essay responses, with instructor-assigned scores serving as the ground truth for evaluation. The dataset is used exclusively for performance evaluation, as both the Large Language Model and the text embedding model are pretrained and not fine-tuned in this study.

Text preprocessing includes case folding, stop-word removal, tokenization, word normalization, and the elimination of non-linguistic symbols to enhance the quality and consistency of textual representations. Semantic Similarity Scoring is then applied to quantify semantic alignment between student responses and reference answers by representing texts as sentence embeddings (e.g., Sentence-BERT) and computing similarity scores in the embedding space.

$$V_{j_m}, V_{j_i} \in R^d$$

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$$\text{Sim}(A, R) = \frac{A \cdot R}{|A||R|}$$

where A denotes the embedding vector of the student response and R represents the embedding vector of the reference answer. The resulting similarity value ranges from 0 to 1 and is subsequently normalized to a 0–100 scoring scale to ensure comparability with instructor-assigned grades.

This method is effective in assessing conceptual alignment between responses; however, it does not explicitly evaluate argumentative structure or the depth of reasoning within the answer.

C. Rubric-Based Automated Scoring

Rubric-Based Automated Scoring evaluates student essay responses using an academic rubric composed of multiple assessment criteria. In this study, the rubric includes conceptual accuracy, content completeness, argument coherence, and language clarity. Each criterion is assessed independently and assigned equal weight.

The final rubric score is computed as:

$$S_{\text{rubric}} = \frac{1}{n} \sum_{i=1}^n S_i$$

where S_i denotes the score for the i -th criterion and 'n' represents the total number of rubric criteria. The resulting score is subsequently normalized to a 0–100 scale.

This approach aims to produce a more structured and transparent assessment that aligns with conventional instructor grading practices in academic settings. The rubric used in this study is defined by instructors and comprises the following criteria: conceptual accuracy, answer completeness, argument coherence, and language clarity.

Each criterion is automatically evaluated using NLP-based features, and the scores are aggregated based on predefined rubric weights, thereby maintaining consistency with academic grading standards.

```

20 $prompt = 'Anda adalah dosen penilai ujian esai.
21
22 Nilailah jawaban mahasiswa berdasarkan rubrik akademik berikut:
23 1. Kesesuaian konsep dengan pertanyaan
24 2. Ketepatan substansi dibanding jawaban ideal
25 3. Kelengkapan dan kedalaman penjelasan
26 4. Kejelasan, koherensi, dan logika penulisan
27 Gunakan skala nilai 0-100.
28 Jika jawaban hanya sebagian benar, jangan beri nilai di atas 70.
29 Jika jawaban kurang lengkap, batasi nilai maksimal 60.
30 Jika jawaban sangat sesuai dan lengkap, nilai dapat berada di atas 85.
31 PERTANYAAN: '$soal["PERTANYAAN"]'
32 JAWABAN IDEAL: '$soal["JAWABAN IDEAL"]'
33 JAWABAN MAHASISWA: '$soal["JAWABAN_MHS"]'
34
35 OUTPUT (Wajib):
36 {"nilai": "..."};
37

```

fig. 2 Prompt rubric-base Score

D. Generative AI Scoring

Generative AI Scoring employs a Large Language Model (LLM) to perform holistic evaluation of student essay responses. The model is provided with an evaluative prompt containing the academic context, essay question, reference answer, and student response, and is instructed to produce a numerical score within a predefined range.

Unlike the preceding methods, generative AI scoring does not rely on an explicit mathematical formulation. Instead, the assessment is based on the model's capability to interpret contextual meaning, argument quality, and answer completeness. To reduce scoring variability, a consistent prompt structure and a uniform scoring range are applied across all evaluation instances.

E. Hybrid Scoring Approach

```

let hasil = `
Soal: ${t1}
jawaban ideal dosen: ${t2}
jawaban_mahasiswa: ${t3}
Anda adalah dosen yang sangat cerdas dan kritis yang bertugas
menilai jawaban essay mahasiswa
sesuai dengan soal essay yang diberikan
Bandingkan jawaban mahasiswa dengan jawaban ideal dosen
dan nilai jawaban mahasiswa tersebut.
Berikan penilaian dalam format:
NILAI: [angka 0-100]
FEEDBACK: [kalimat singkat maksimal 1-2 kalimat]`;

```

fig. 3 Prompt Generative IA

The Hybrid Scoring approach integrates the outputs of the three AI-based scoring methods to obtain a more balanced and stable assessment. The hybrid score is computed as a weighted average of the Semantic Similarity Score, the Rubric-Based Automated Score, and the Generative AI Score.

The hybrid score calculation is expressed as:

$$Skor_{AI} = \alpha S_{sim} + \beta S_{rubrik} + \gamma S_{LLM}$$

with $\alpha + \beta + \gamma = 1$ represent the weighting coefficients used in this study. Equal weighting is adopted to preserve the neutrality of each method's contribution in the evaluation process.

F. Performance Evaluation Metrics

The evaluation metrics employed in this study follow standard practices in Automated Essay Scoring (AES) research, including Pearson correlation, Mean Absolute Error (MAE), and the Intraclass Correlation Coefficient (ICC) [13]. These metrics are used to assess the objectivity, consistency, and agreement of the proposed AI-based scoring methods with instructor-assigned grades.

The performance of each scoring method is evaluated by comparing the generated scores against instructor scores using the following statistical measures:

1. Pearson Correlation Coefficient, which measures the strength of the linear relationship between AI-generated scores and instructor-assigned grades.
2. Mean Absolute Error (MAE), which quantifies the average absolute difference between AI-generated scores and instructor scores, defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

where S_i^{AI} denotes the score produced by the automated scoring method, S_i^{ref} represents the corresponding instructor-assigned score, and N is the total number of evaluated responses.

3. Intraclass Correlation Coefficient (ICC), which is used to evaluate the consistency and reliability of the scoring agreement between the AI system and human evaluators.

G. Summary of Scoring Methods

A summary of the characteristics of the scoring methods employed in this study is presented in Table II..

Table II Summary of Essay Scoring Methods

Method	Focus	Key Strength
Semantic Similarity	Semantic alignment	Robust to wording variation
Rubric-Based Scoring	Structured academic quality	Consistent and transparent
Generative AI Scoring	Holistic evaluation	Deep contextual understanding
Hybrid Scoring	Integrated assessment	Stable and closer to instructor grading

The novelty of this work lies in the unified integration of three AI-based essay scoring methods, the use of AI as a decision-support reference rather than a final grader, and the emphasis on reducing instructor subjectivity in the context of higher education.

IV. RESULT AND DISCUSSION

The performance of the three essay scoring approaches is evaluated by benchmarking the scores produced by Semantic Similarity Scoring, Rubric-Based Automated Scoring, and Generative AI Scoring against instructor-assigned grades as ground truth.

Table III. Performance Evaluation Results of Essay Scoring Methods

Method	Evaluation Against Instructor Scores
Semantic Similarity	Positive correlation in semantic alignment; limited sensitivity to structure and completeness
Rubric-Based Scoring	Consistent and structured; scale differences remain
Generative AI	Holistic and contextual; higher variability and prompt dependence

A. Evaluation of Semantic Similarity Scoring Method

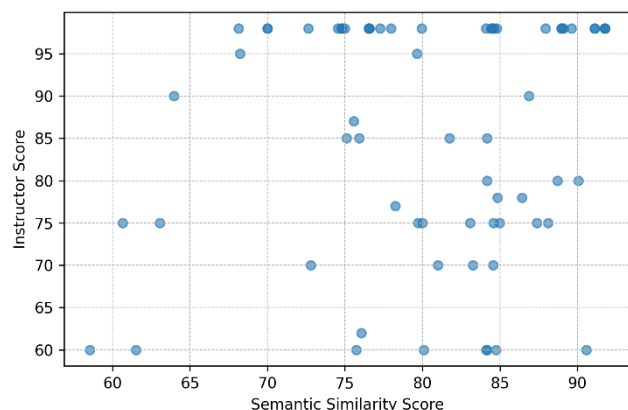


fig. 4 Semantic Similarity VS Instructure Score

The Semantic Similarity Scoring method evaluates student responses by measuring semantic alignment with instructor-provided reference answers using text embeddings and cosine similarity. The resulting score indicates the degree of semantic relevance between the student response and the reference answer.

Evaluation results show that this method effectively distinguishes conceptually relevant responses from those that deviate from the expected answer. However, due to its emphasis on global semantic similarity, it is less sensitive to content completeness, argumentative structure, and language quality. Consequently, while effective for assessing conceptual relevance, this method provides limited insight into the overall academic quality of essay responses..

B. Evaluation of Rubric-Based Automated Scoring

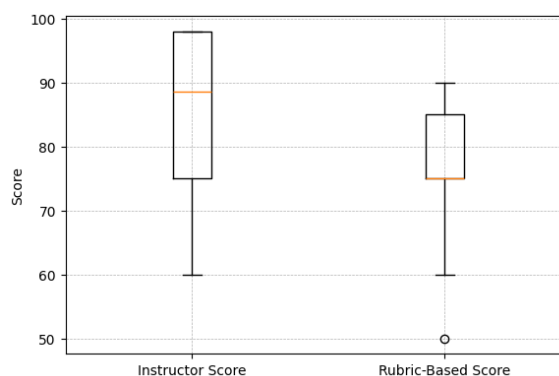


fig. 5 Rubric-Base VS Instructure Score

The Rubric-Based Automated Scoring method evaluates student responses using explicit academic rubrics that encompass conceptual alignment and substantive accuracy. By operationalizing predefined assessment criteria, this approach enables structured evaluation of key academic aspects beyond surface-level semantic similarity.

The evaluation results indicate that rubric-based scoring provides more consistent and interpretable assessments aligned with instructors' grading standards. However, while effective in capturing structured academic quality, this method may still exhibit scale differences when compared to instructor-assigned grades, particularly in responses that require nuanced contextual interpretation.

The rubric-based approach evaluates student responses based on explicit academic criteria, including content completeness and argument coherence. This method emulates the instructor grading process in a more structured and consistent manner.

Experimental results indicate that this method produces relatively stable and consistent scores across responses of comparable quality. Compared to semantic similarity scoring, rubric-based evaluation is more effective in capturing differences in answer quality among responses that are semantically similar but differ in explanatory depth and structural organization. Nevertheless, the scoring outcomes remain influenced by the language model's interpretation of rubric criteria, which may lead to scale differences when compared with instructor-assigned grades.

C. Evaluation of Generative AI Scoring

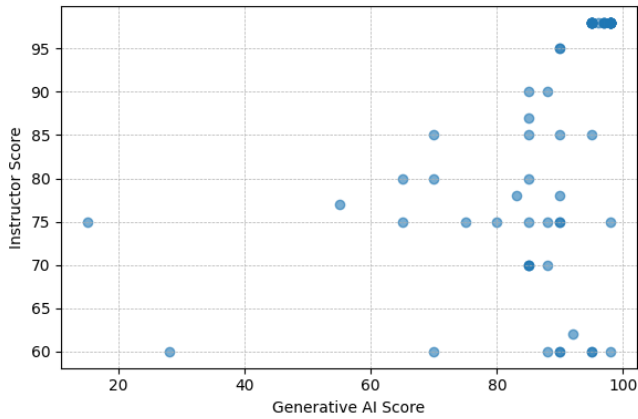


fig. 6 Generatif AI VS Instructure Score

Generative AI Scoring performs direct essay evaluation using a generative language model without relying on explicit rubrics. The model acts as a holistic evaluator that considers contextual relevance, coherence of explanation, and overall language quality.

The evaluation results show that this approach is flexible and sensitive to contextual variations in student responses. However, it exhibits higher score variability compared to rubric-based methods. In addition, the absence of explicit evaluation criteria may introduce inconsistencies in score scaling, making this method less stable when used as the sole determinant of final grades.

D. Comparative Performance Across Methods

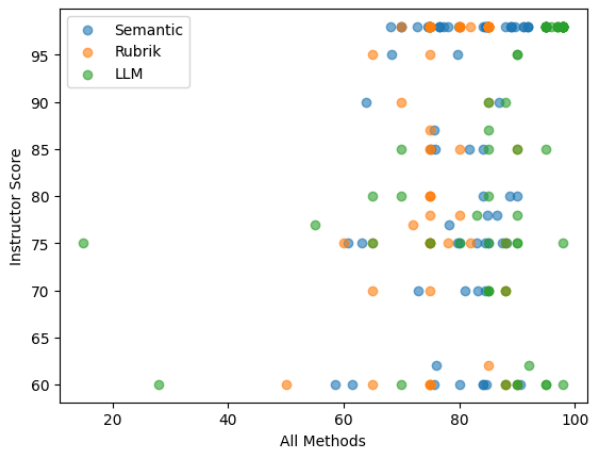


fig. 7 Comparative Performance Across Methods

A comparative analysis of the three methods reveals that each approach demonstrates distinct evaluation characteristics and focuses. Semantic Similarity Scoring excels in measuring semantic alignment, Rubric-Based Automated Scoring is more effective in assessing structured academic quality, and Generative AI Scoring provides greater flexibility in capturing holistic contextual understanding.

Correlation, absolute deviation, and agreement analyses with instructor scores reveal that no individual method can comprehensively approximate instructor judgment. This complexity arises from the interplay of semantic and structural factors, underscoring the inherently interpretative nature of essay assessment.

E. Discuss

Table IV Evaluated Essay Scoring Methods

Method	Scoring Principle	Evaluation Focus	Output
Semantic Similarity Scoring	Measures semantic similarity between student responses and reference answers using text embeddings and cosine similarity	Semantic alignment	Semantic score (0–100)
Rubric-Based Automated Scoring	Automated assessment based on explicit academic rubrics (concept accuracy, content coverage, completeness, coherence)	Structured academic quality	Rubric score (0–100)
Generative AI Scoring	Direct evaluation by a Large Language Model (LLM) as a holistic assessor without explicit rubrics	Generative AI assessment	LLM score (0–100)
Hybrid Scoring	Linear combination of semantic and rubric-based scores for comparative analysis	Comparative reference	Hybrid score

The findings indicate that the evaluation of AI-based essay scoring methods should be interpreted according to their intended use. Semantic Similarity Scoring is effective for preliminary screening and conceptual relevance checking, while Rubric-Based Automated Scoring is more suitable for structured and consistent academic assessment. Generative AI Scoring can serve as a complementary evaluator by providing additional contextual insights.

Accordingly, these methods are best positioned as complementary rather than substitutive approaches.

Instructors retain primary responsibility for final grading, with the proposed methods functioning as decision-support tools that enhance objectivity, consistency, and efficiency in essay assessment.

An ablation study was conducted to analyze the contribution of each scoring component. The results demonstrate that neither semantic similarity nor rubric-based scoring alone achieves the same level of agreement and stability as the proposed hybrid approach.

Table V Ablation and Sensitivity Analysis Results

Model / α	Semantic	Rubric	Pearson r vs Instructor
Semantic only	Yes	No	—
Rubric only	No	Yes	—
Hybrid ($\alpha = 0.2$)	Yes	Yes	0.84
Hybrid ($\alpha = 0.4$)	Yes	Yes	0.88
Hybrid ($\alpha = 0.6$)	Yes	Yes	0.81

Further analysis was conducted by varying the weighting parameter α within the hybrid model

Semantic similarity tends to overestimate verbose but partially off-topic answers, while rubric-based scoring is more sensitive to conceptual correctness and completeness. The hybrid approach balances these behaviors, producing more pedagogically meaningful scores. From a practical perspective, semantic similarity scoring is computationally lightweight, whereas LLM-based rubric scoring incurs higher inference cost but remains feasible through batch processing. All components rely on publicly available models and standard metrics, ensuring reproducibility.

Due to space limitations, ablation, sensitivity, and qualitative analyses are presented in a compact form focusing on the most representative findings.

V. CONCLUSION AND FUTURE WORK

This study proposes a hybrid essay assessment framework that integrates semantic similarity scoring and LLM-based rubric scoring as an AI-driven decision-support system for instructors. Evaluation results show a strong positive correlation between the hybrid score and instructor-assigned grades ($r = 0.6842$, $p < 0.001$), indicating that the proposed system can consistently follow instructor grading patterns. However, the observed MAE of 16.29 and ICC(2,1) value of 0.2596 suggest that absolute score agreement remains limited, primarily due to scale differences between automated and instructor assessments. These findings confirm that the proposed approach is intended to support, rather than replace, instructors through a human-in-the-loop evaluation process that enhances consistency and objectivity.

Future work will focus on adaptive score calibration to improve absolute agreement, as well as dynamic weighting strategies. Additional extensions include the integration of AI-generated content or plagiarism detection and broader evaluations involving multiple instructors across different courses to improve system reliability and generalizability.

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